

Data Sources and Data Preparation for SSNbler and SSN2

**Dave Nagel
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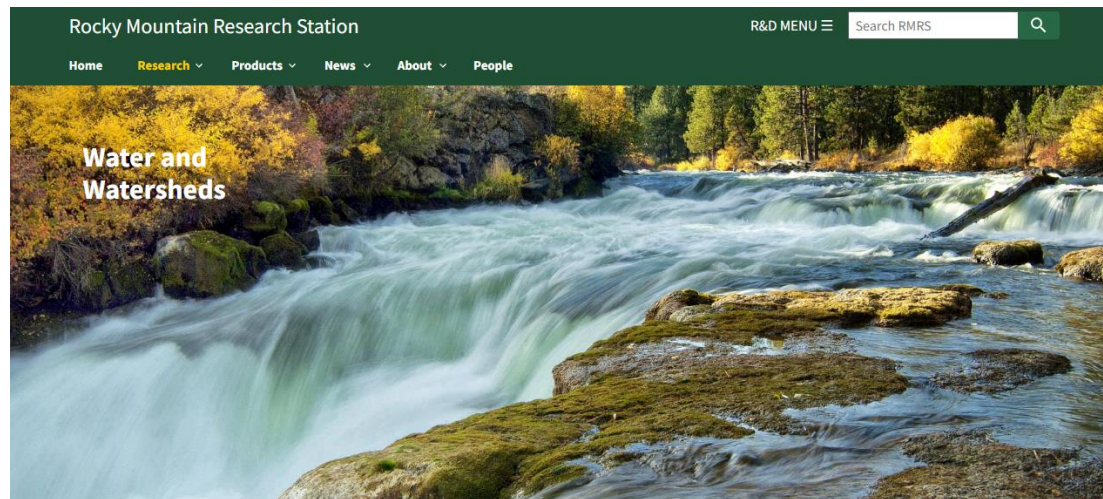
**U.S. Forest Service, Rocky Mountain Research Station
Water and Watersheds Program**

Ver. September 24, 2026

US Forest Service

Rocky Mountain Research Station

Water and Watersheds Program



**Google: Water and Watersheds US
Forest Service**

<https://research.fs.usda.gov/rmrs/programs/waterwatersheds>

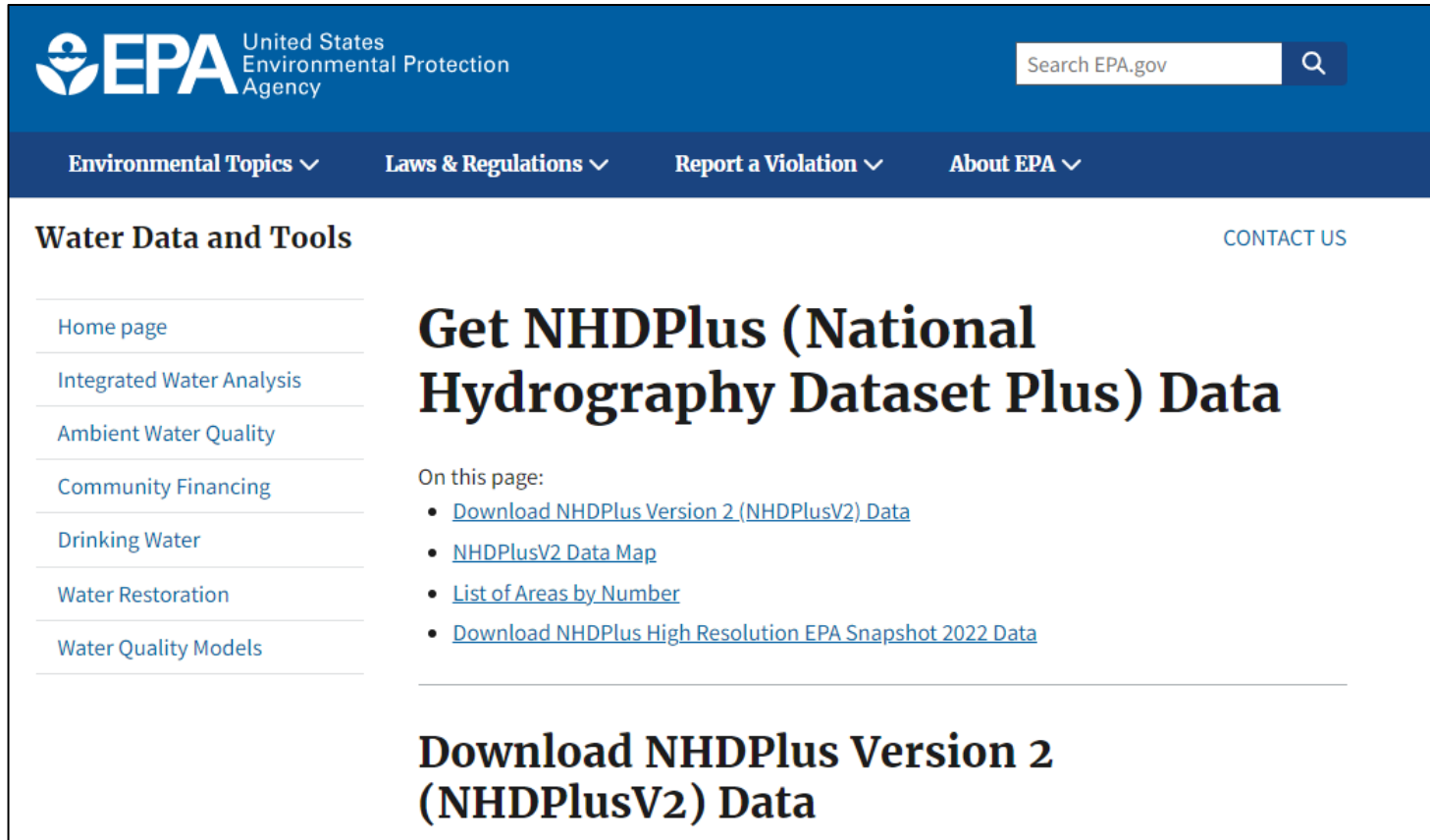
Data Sources

- 1) NHDPlusV2**
- 2) National Stream Internet (NSI)**
- 3) StreamCat**
- 4) NorWeST**

Data Preparation for SSNbler and SSN2

- 1) Input data**
- 2) Managing IDs**
- 3) Covariates**
- 4) Input to SSNbler**

NHDPlusV2



The screenshot shows the EPA website's 'Water Data and Tools' section. The header includes the EPA logo and navigation links. The main content area features a sidebar with links to various water data tools and a main section titled 'Get NHDPlus (National Hydrography Dataset Plus) Data'. This section lists four links: 'Download NHDPlus Version 2 (NHDPlusV2) Data', 'NHDPlusV2 Data Map', 'List of Areas by Number', and 'Download NHDPlus High Resolution EPA Snapshot 2022 Data'. Below these links is a large heading for 'Download NHDPlus Version 2 (NHDPlusV2) Data'.

EPA United States Environmental Protection Agency

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Water Data and Tools [CONTACT US](#)

Home page

Integrated Water Analysis

Ambient Water Quality

Community Financing

Drinking Water

Water Restoration

Water Quality Models

Get NHDPlus (National Hydrography Dataset Plus) Data

On this page:

- [Download NHDPlus Version 2 \(NHDPlusV2\) Data](#)
- [NHDPlusV2 Data Map](#)
- [List of Areas by Number](#)
- [Download NHDPlus High Resolution EPA Snapshot 2022 Data](#)

Download NHDPlus Version 2 (NHDPlusV2) Data

Google: NHDPlusV2 EPA

<https://www.epa.gov/waterdata/get-nhdplus-national-hydrography-dataset-plus-data>

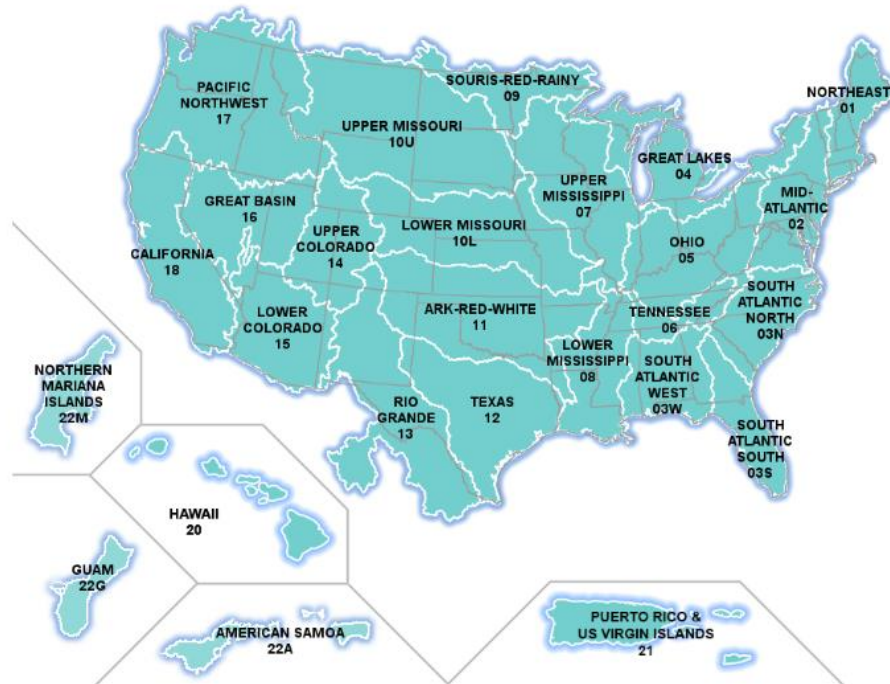
NHDPlusV2

1:100,000 Scale National Streamlines



NHDPlusV2

[National Data](#) | [Global Data](#)

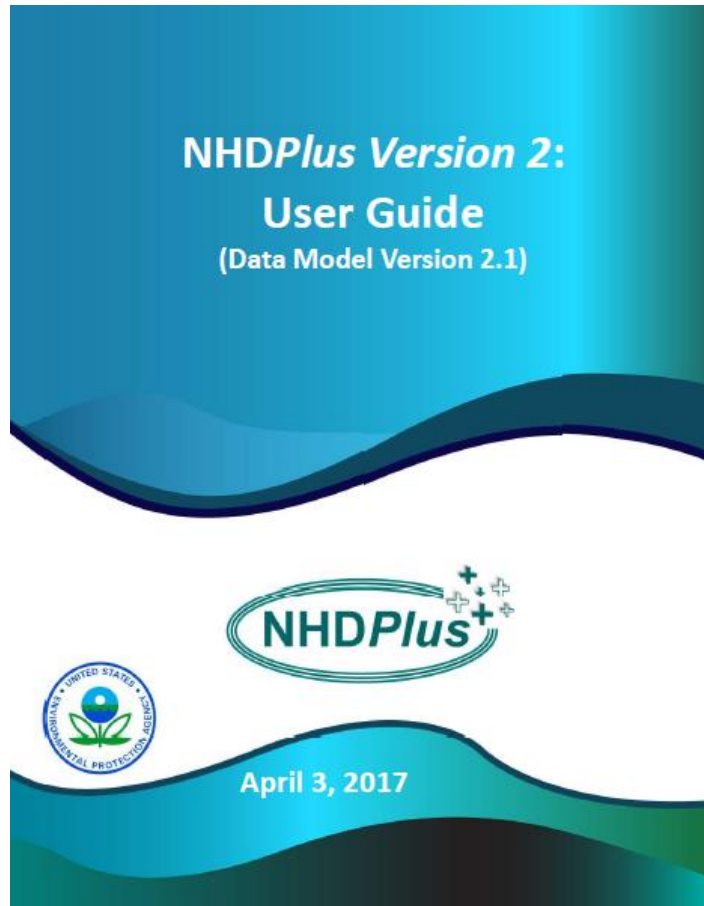


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[NHDPlusV21_PN_17_NHDSnapshot_08.7z](#) (102.1 MB)

NHDPlusV2



Data

- DEM
- Catchments
- Watershed boundaries

Themes/Covariates

- Drainage area
- Slope
- Flow (Discharge)
- Velocity
- Precipitation
- Stream order
- Etc.

National Stream Internet NSI

**Topologically Corrected NHDPlusV2 Data For Use With
SSNbler and SSN2**

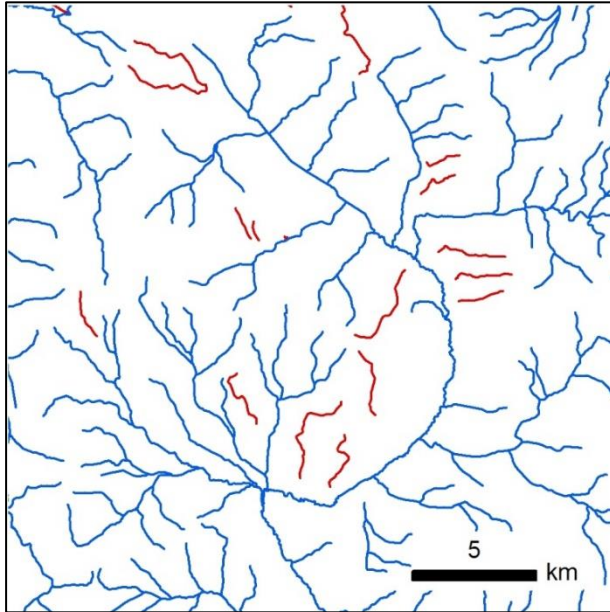


**Google: National Stream Internet US Forest
Service**

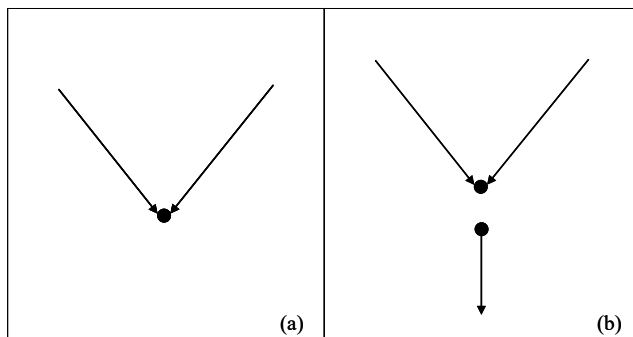
<https://research.fs.usda.gov/rmrs/projects/national-stream-internet#overview>

NSI Edits

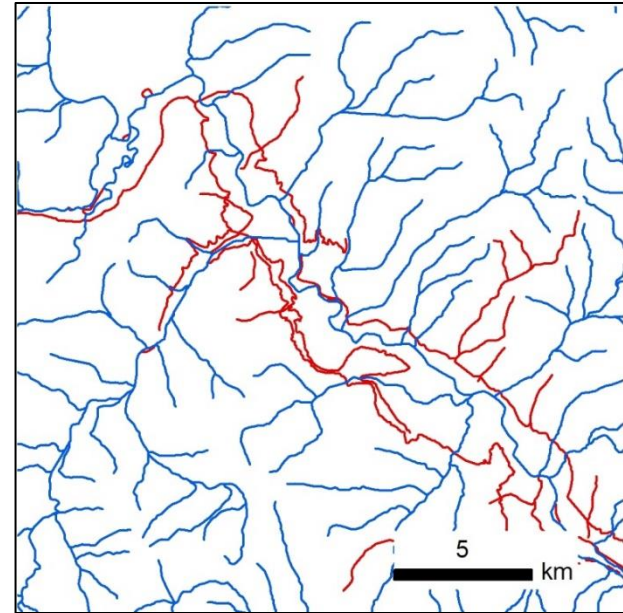
Uninitialized Flow



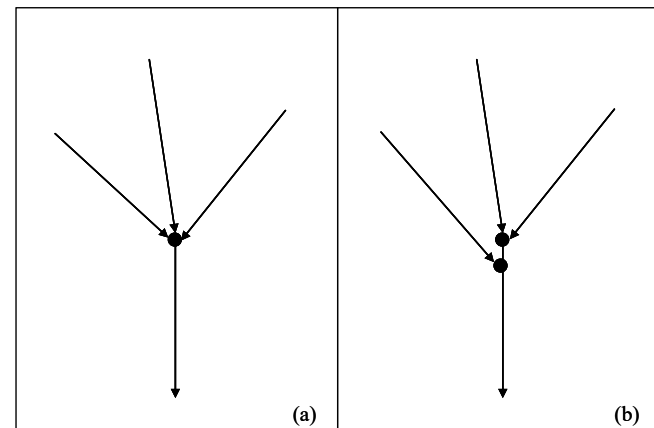
Converging Streams



Braids and Diversions



Complex Confluences



National Stream Internet NSI

National Stream Internet Protocol and User Guide

David Nagel¹, Erin Peterson², Daniel Isaak¹, Michael Dumelle³, Jay Ver Hoef⁴, and
Dona Horan¹

USDA Forest Service, Rocky Mountain Research Station
Water and Watersheds Program
322 E. Front St., Boise, ID



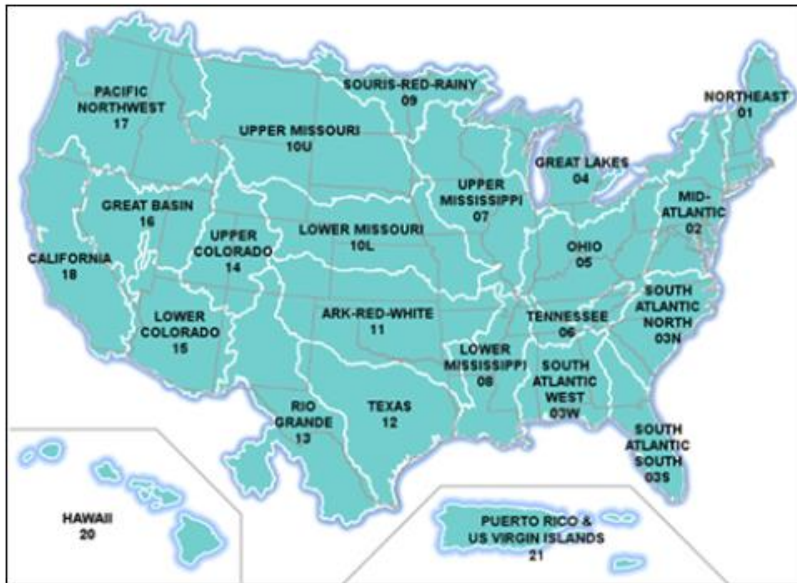
UNITED STATES DEPARTMENT OF AGRICULTURE - U.S. FOREST SERVICE



Rocky Mountain Research Station
Water and Watersheds Program



National Stream Internet NSI



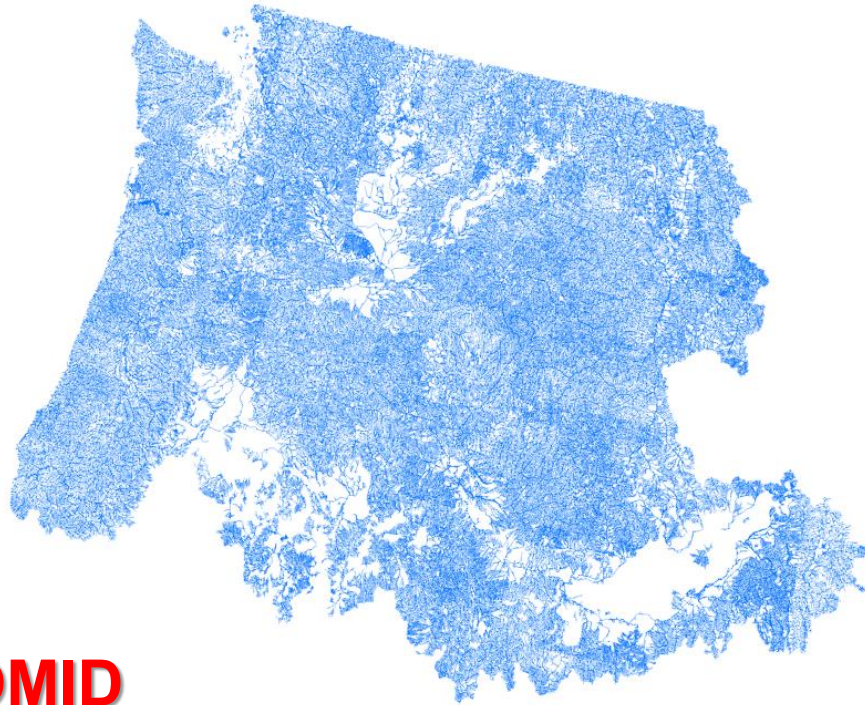
Download from the main page
Organized by region

17 Pacific Northwest

[Flowline](#)

[Prediction Points](#)

National Stream Internet NSI



**PNW Region 17
streamlines**

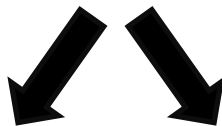
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2	1	Polyline	23123523	9/6/2005	Medium	1533479	Snake River	0.294	17040101000004	With Digitized	ArtificialPath	55800	0.3204	1365.345
3	2	Polyline	23123853	2/7/2001	Medium			1.496	17040101000012	With Digitized	ArtificialPath	55800	2.3715	2.3715
4	3	Polyline	23123851	2/7/2001	Medium			0.448	17040101000013	With Digitized	ArtificialPath	55800	0.8991	0.8991
5	4	Polyline	23123879	2/7/2001	Medium			0.314	17040101000014	With Digitized	ArtificialPath	55800	0.4842	0.4842
6	5	Polyline	23123969	2/7/2001	Medium			0.247	17040101000015	With Digitized	ArtificialPath	55800	0.3447	0.3447
7	6	Polyline	23123979	2/7/2001	Medium			0.379	17040101000016	With Digitized	ArtificialPath	55800	1.0017	1.0017
8	7	Polyline	23124021	2/7/2001	Medium			0.477	17040101000017	With Digitized	ArtificialPath	55800	1.2717	1.2717
9	8	Polyline	23124049	2/7/2001	Medium			0.423	17040101000018	With Digitized	ArtificialPath	55800	1.7073	3.8727

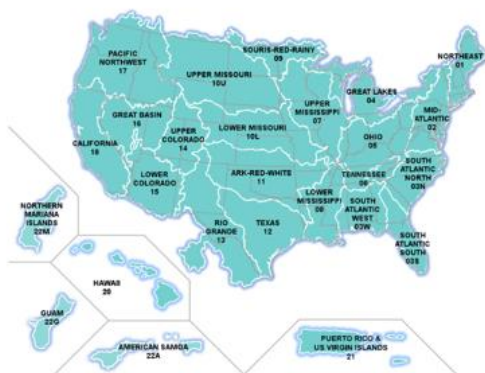
COMID

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1	0	Polyline	23123525	9/6/2005	Medium	1533479	Snake River	0.292	17040101000004	With Digitized	ArtificialPath	55800	0.3384	1365.0246
2	1	Polyline	23123523	9/6/2005	Medium	1533479	Snake River	0.294	17040101000004	With Digitized	ArtificialPath	55800	0.3204	1365.345
3	2	Polyline	23123853	2/7/2001	Medium			1.496	170401010000012	With Digitized	ArtificialPath	55800	2.3715	2.3715
4	3	Polyline	23123851	2/7/2001	Medium			0.448	170401010000013	With Digitized	ArtificialPath	55800	0.8991	0.8991
5	4	Polyline	23123879	2/7/2001	Medium			0.314	170401010000014	With Digitized	ArtificialPath	55800	0.4842	0.4842
6	5	Polyline	23123969	2/7/2001	Medium			0.247	170401010000015	With Digitized	ArtificialPath	55800	0.3447	0.3447
7	6	Polyline	23123979	2/7/2001	Medium			0.379	170401010000016	With Digitized	ArtificialPath	55800	1.0017	1.0017
8	7	Polyline	23124021	2/7/2001	Medium			0.477	170401010000017	With Digitized	ArtificialPath	55800	1.2717	1.2717
9	8	Polyline	23124049	2/7/2001	Medium			0.423	170401010000018	With Digitized	ArtificialPath	55800	1.7073	3.8727

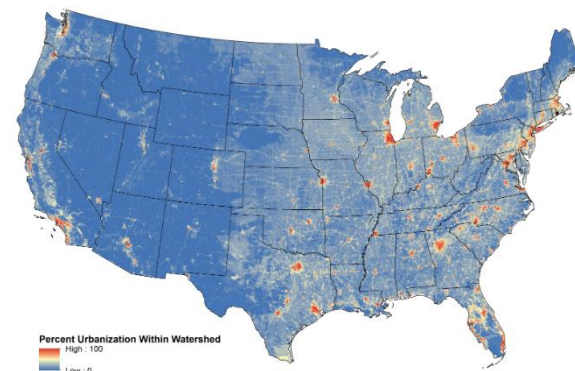
COMID



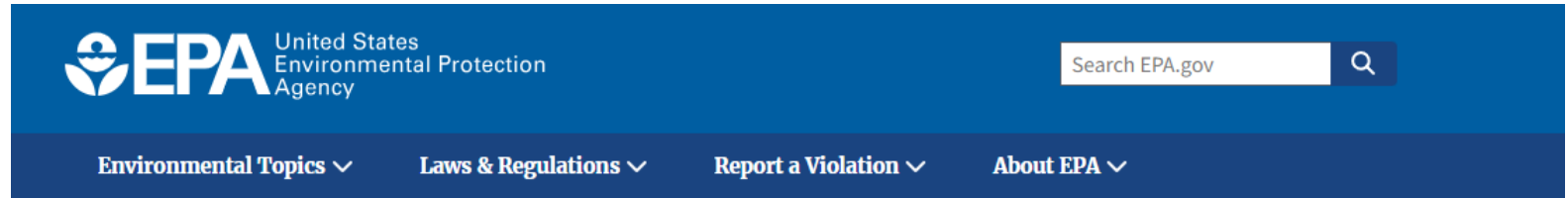
NHDPlusV2

[illegible]

StreamCat



StreamCat Dataset



National Aquatic Resource Surveys

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[Background](#)

[Indicators](#)

[Manuals](#)

StreamCat Dataset

On this page:

[Access StreamCat Data](#) | [StreamCat Documentation](#) | [Data Structure](#) | [Feedback](#)

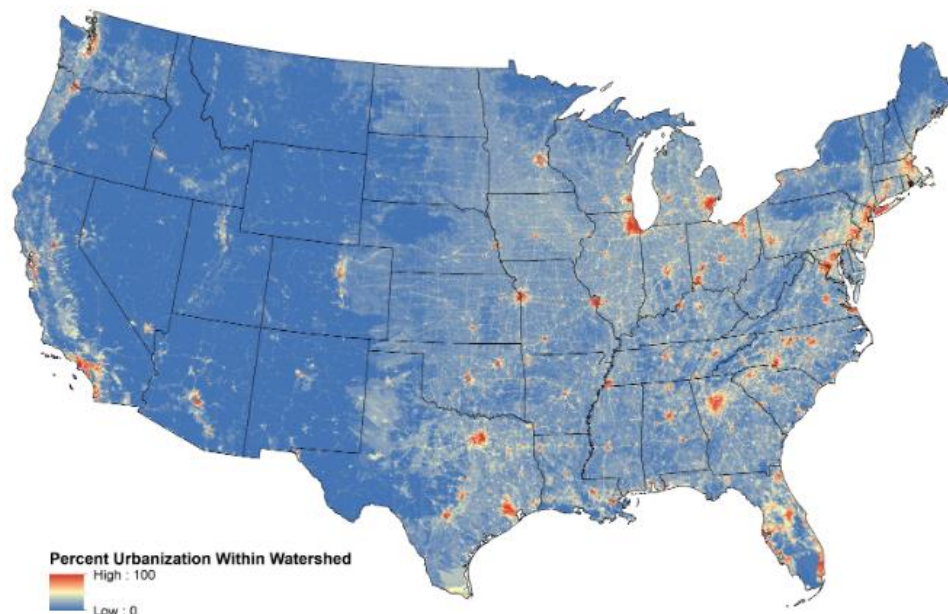
Google: StreamCat EPA

<https://www.epa.gov/national-aquatic-resource-surveys/streamcat-dataset>

StreamCat Dataset

132 Metrics

- Base flow index
- Elevation
- Dams
- Forest loss
- Fire perimeters
- Lithology
- Land cover
- Road density
- Precipitation
- Etc.



<https://www.epa.gov/national-aquatic-resource-surveys/streamcat-dataset>

StreamCat Access Options

Access StreamCat Data

- Data by Individual Catchments via [Map Interface](#)
- Data by Regions, States, Counties via [Menu Interface](#) 
- [R Programming Interface](#)  [↗](#)
- Data via [EPA WATERS Watershed Characterization Service](#)
 - This is a service providing an HTML report or JSON object listing both standard NHDPlus and StreamCat metrics for a given COMID provided by the user. See HTML Examples in link above, and simply pass a COMID of interest to end of the URL (i.e. pComID=4795168).

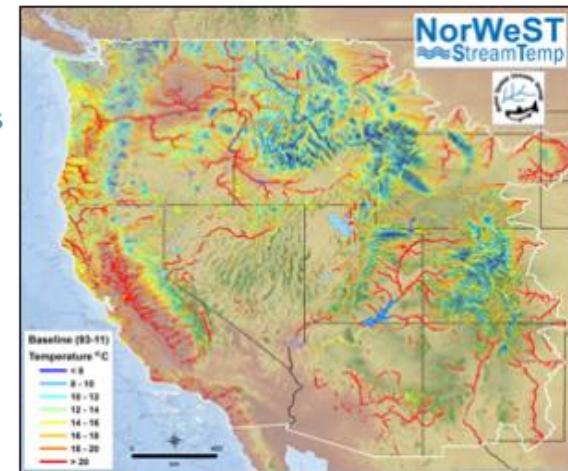
NorWeST Stream Temperature

NorWeST

Status: Ongoing

The NorWeST project page hosts stream temperature data and climate scenarios in a variety of user-friendly digital formats for streams and rivers across the western U.S. The temperature database was compiled from hundreds of biologists and hydrologists working for >100 resource agencies and contains >200,000,000 hourly temperature recordings at >20,000 unique stream sites. Those temperature data were used with spatial statistical network models to develop 36 historical and future climate scenarios at 1-kilometer resolution for >1,000,000 kilometers of stream.

Temperature data and model outputs, registered to NHDPlus stream lines, are posted to the website after QA/QC procedures and development of the final temperature model within a river basin. It is hoped that open access to the data and the availability of accurate stream temperature scenarios will foster new research and collaborative relationships that enhance management and conservation of aquatic resources.



Google: NorWeST US Forest Service Research

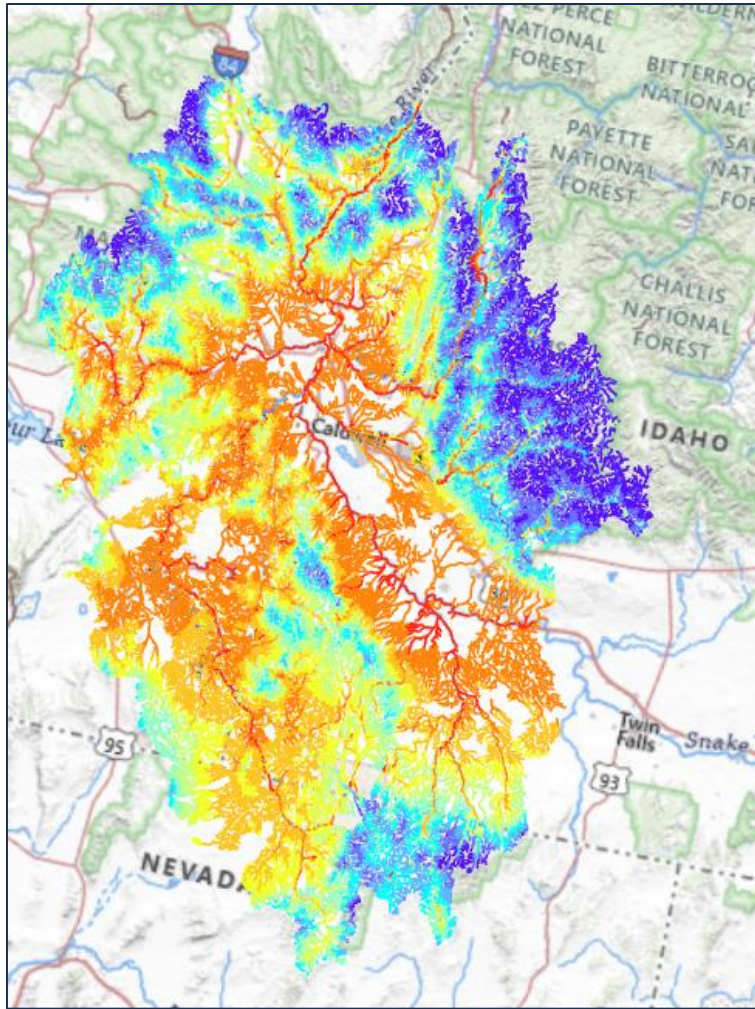
<https://research.fs.usda.gov/rmrs/projects/norwest>

NorWeST Stream Temperature

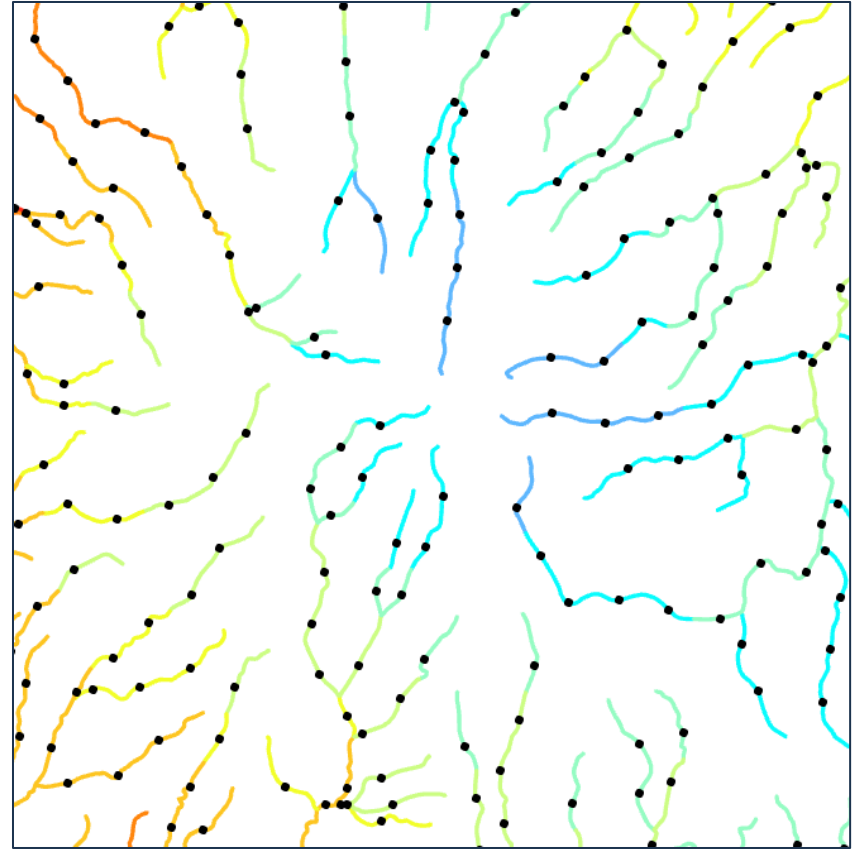
NorWeST Processing Units



NorWeST Stream Temperature



Middle Snake

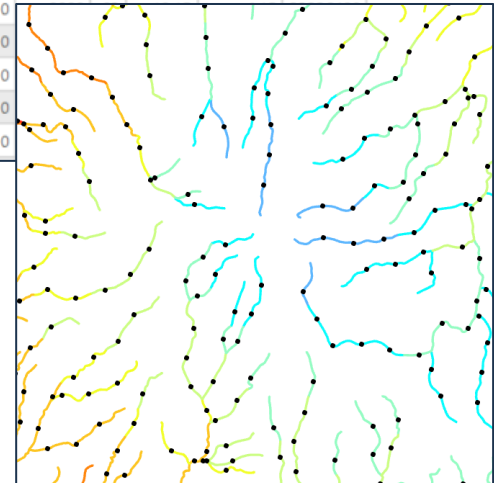


**1 km reaches and
prediction points**

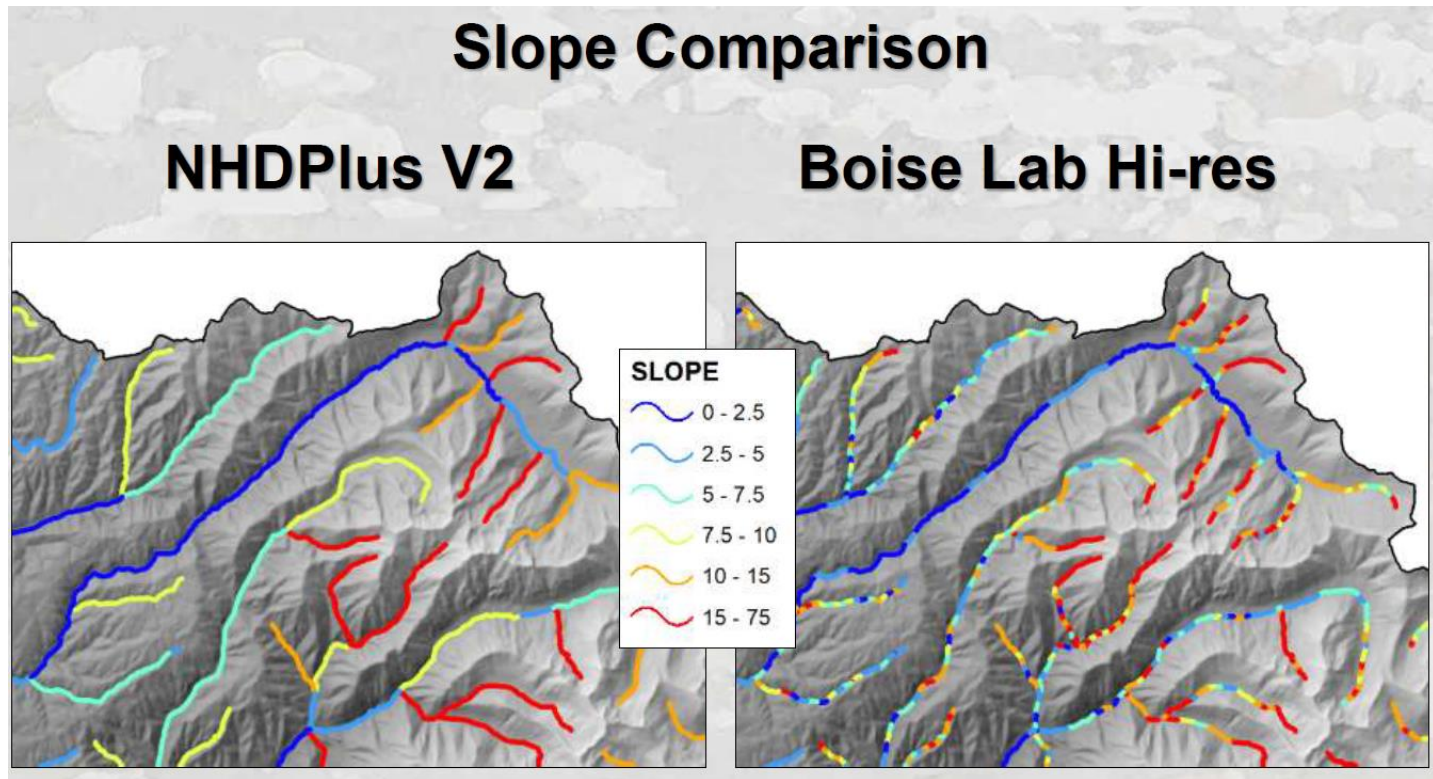
NorWeST Stream Temperature

Attributes

FID	Shape *	OBSPREID_ID	FTYPE	WATERBODY	TAILWATER	ELEV	CANOPY	SLOPE	PRECIP	CUMDRAINAG	Y_COORD	NLCD11PC	GLACIER	BFI	Air_Aug	Flow_Aug	COMID	GNIS_NAME	S1_93_11
0	Polyline	100001	StreamRiver	0	0	2133.57	7.89	0.05316	427.2545	4.24	1223893.22	0	-9999	63	19.15	4.15	23336510		13.36
1	Polyline	100002	StreamRiver	0	0	2085.49	4.04	0.05316	427.2545	4.24	1224757.21	0	-9999	63	19.15	4.15	23336510		13.67
2	Polyline	100003	StreamRiver	0	0	2064.6	0	0.00807	405.4112	0.71	1225083.34	0	-9999	63	19.15	4.15	23336504		14.23
3	Polyline	100004	StreamRiver	0	0	2023.54	9.72	0.03673	389.1128	26.53	1225197.45	0	-9999	63	19.15	4.15	23336496	Indian Creek	14.27
4	Polyline	100005	StreamRiver	0	0	1998.55	19.18	0.03673	389.1128	26.53	1225743.96	0	-9999	63	19.15	4.15	23336496	Indian Creek	14.26
5	Polyline	100006	StreamRiver	0	0	1961.81	22.84	0.03673	389.1128	26.53	1226642.26	0	-9999	63	19.15	4.15	23336496	Indian Creek	14.4
6	Polyline	100007	StreamRiver	0	0	1913.36	14.65	0.03673	389.1128	26.53	1227500.41	0	-9999	63	19.15	4.15	23336496	Indian Creek	14.78
7	Polyline	100008	StreamRiver	0	0	2061.26	5	0.01334	404.72	18.05	1225250.68	0	-9999	63	19.15	4.15	23336500	Indian Creek	14.32
8	Polyline	100009	StreamRiver	0	0	2172.69	4.08	0.02837	435.7138	8.25	1226623.55	0	-9999	63	19.15	4.15	23336502	Indian Creek	13.53
9	Polyline	100010	StreamRiver	0	0	2141.23	13.7	0.02837	435.7138	8.25	1225961.4	0	-9999	63	19.15	4.15	23336502	Indian Creek	13.57
10	Polyline	100011	StreamRiver	0	0	2120.56	23.38	0.02837	435.7138	8.25	1225507.91	0	-9999	63	19.15	4.15	23336502	Indian Creek	13.54
11	Polyline	100012	StreamRiver	0	0	2102.89	13.48	0.02837	435.7138	8.25	1225115.46	0	-9999	63	19.15	4.15	23336502	Indian Creek	13.79
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13	Polyline	100014	StreamRiver	0	0	2072.84	11.7	0.0068	404.72	13.32	1225356.59	0	-9999	63	19.15	4.15	23336488	Indian Creek	14.21
14	Polyline	100015	StreamRiver	0	0	1985.77	1.94	0.04199	396.4815	3.31	1227597.05	0	-9999	63	19.15	4.15	23336484		14.35
15	Polyline	100016	StreamRiver	0	0	2131.69	1.15	0.0941	423.7314	1.8	1227765.08	0	-9999	63	19.15	4.15	23336480		13.13
16	Polyline	100017	StreamRiver	0	0	2050.14	11.17	0.0941	423.7314	1.8	1228587.47	0							13.42
17	Polyline	100018	StreamRiver	0	0	1962.51	12.44	0.0619	417.7257	4.25	1229401.42	0							14.16
18	Polyline	100019	StreamRiver	0	0	1912.58	5.85	0.0619	417.7257	4.25	1230303.7	0							14.52
19	Polyline	100020	StreamRiver	0	0	1989.55	8.17	0.05252	359.5108	6.48	1227867.62	0							14.17
20	Polyline	100021	StreamRiver	0	0	1926.6	19.67	0.05252	359.5108	6.48	1228696.86	0							14.33



High Resolution Stream Gradient for NHDPlusV2



**Unpublished dataset for the
Western U.S. on external drive**

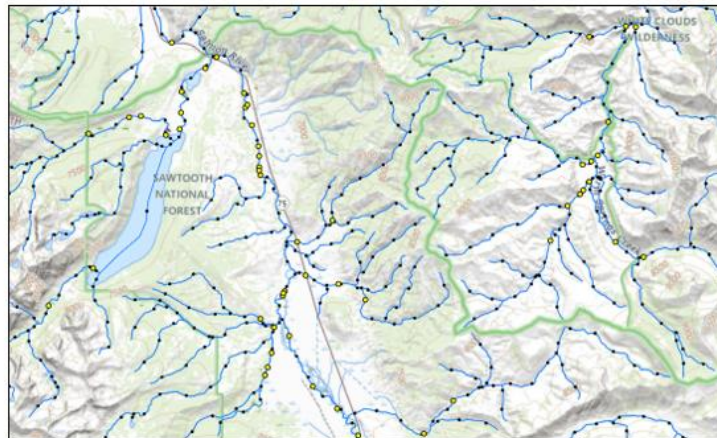
Preparing Input Data for SSNbler and SSN2

Preparing Input Data for Spatial Stream Network Modeling

David Nagel¹ and Erin Peterson²

U.S. Forest Service, Rocky Mountain Research Station
Water and Watersheds Program
Boise Aquatic Science Laboratory
322 E. Front St., Boise, ID

May 6, 2024



Document:

<https://research.fs.usda.gov/rmrs/understory/preparing-input-data-spatial-stream-network-modeling>

Preparing Input Data for SSNbler and SSN2

1) Gather streamlines, observation sites, and prediction points



Tip: Snap observation sites to streams

Pro: Editing Tools > Snap

Preparing Input Data for SSNbler and SSN2

2) Check topology and make corrections if necessary for the streamline feature class

Correcting topological errors using
SSNbler and QGIS

Erin E Peterson and Alan Pearse
May 6, 2024

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Correcting topological errors using
SSNbler and ArcGIS Pro

David Nagel and Erin E Peterson
May 2, 2024

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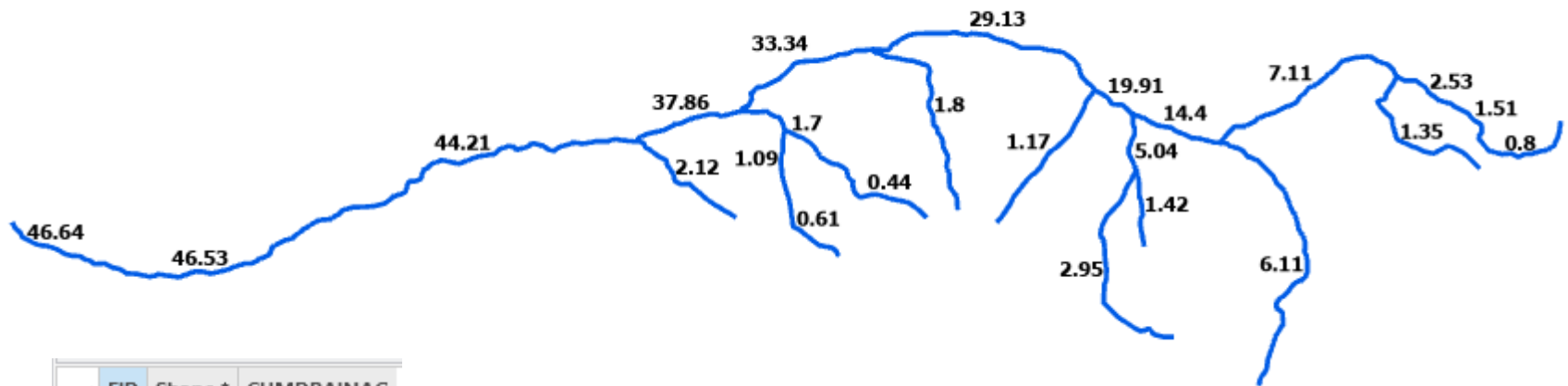
1. Introduction

To generate a valid landscape network (LSN) with the SSNbler R package, users must ensure their digitised streams are free of topological errors and topological restrictions not permitted in SSN2. Vector streams data are sometimes prone to topological errors, even from validated public domain sources such as USGS. The aim of this tutorial is to illustrate the common topological errors sometimes found in digital stream networks and provide guidance on correcting those. It has been written for SSNbler v0.1.0 and assumes the user has some familiarity with ArcGIS Pro and a good working knowledge of R Statistical Software version $\geq 4.0.0$ software.

TopologyEditing_QGIS.pdf

SSNbler_TopologyEditing_Pro.pdf

Preparing Input Data for SSNbler and SSN2 – Streamlines



	FID	Shape *	CUMDRAINAG
1	0	Polyline	6.11
2	1	Polyline	6.11
3	2	Polyline	6.11
4	3	Polyline	6.11
5	4	Polyline	2.95
6	5	Polyline	2.95
7	6	Polyline	2.95
8	7	Polyline	0.61
9	8	Polyline	1.09

3) Assign the weighting variable, such as cumulative drainage area, to every stream reach

Preparing Input Data for SSNbler and SSN2 – Observation Sites

**4) Assign
OBSPRED_ID and
SITE_ID to
observation sites**

**SITE_ID is the location
ID**

	FID	Shape *	OBSPRED_ID	SITE_ID	SampleYear	MeanAugT
1	0	Point	1	303	2001	9.107419
2	1	Point	2	303	2002	12.709355
3	2	Point	3	303	2003	8.77129
4	3	Point	4	303	2004	8.351724
5	4	Point	5	303	2006	8.603793
6	5	Point	6	303	2007	8.955161
7	7	Point	8	303	2012	8.197742
8	8	Point	9	303	2014	8.603548
9	9	Point	10	303	2020	9.592667
10	10	Point	11	9402	1994	11.782903
11	11	Point	12	9402	2001	14.899032
12	12	Point	13	9402	2002	9.290323
13	13	Point	14	9402	2004	9.830667
14	14	Point	15	9402	2014	9.257419

Preparing Input Data for SSNbler and SSN2



**Site 303 has 9 points with the same
XY coordinate**

Preparing Input Data for SSNbler and SSN2 – Prediction Points

**5) Assign
OBSPRED_ID to
prediction points.
Begin numbering at
100,000.**

FID	Shape *	OBSPRED_ID
0	Point	100000
1	Point	100001
2	Point	100002
3	Point	100003
4	Point	100004
5	Point	100005
6	Point	100006
7	Point	100007
8	Point	100008
9	Point	100009
10	Point	100010
11	Point	100011
12	Point	100012

Preparing Input Data for SSNbler and SSN2

6) Merge observation sites and prediction points into a single feature class



	OBJECTID *	Shape *	OBSPRED_ID	SITE_ID	SampleYear	MeanAugT
27	27	Point	27	11191	2015	9.525484
28	28	Point	28	11548	2000	11.640968
29	29	Point	29	39218	2014	8.787097
30	30	Point	30	39244	2009	7.94871
31	31	Point	31	39244	2014	7.955806
32	32	Point	32	39245	2014	8.316129
33	33	Point	33	39261	2015	10.104516
34	34	Point	34	39277	2013	8.899032
35	35	Point	35	39277	2014	8.546129
36	36	Point	100000	<Null>	<Null>	<Null>
37	37	Point	100001	<Null>	<Null>	<Null>
38	38	Point	100002	<Null>	<Null>	<Null>
39	39	Point	100003	<Null>	<Null>	<Null>
40	40	Point	100004	<Null>	<Null>	<Null>
41	41	Point	100005	<Null>	<Null>	<Null>
42	42	Point	100006	<Null>	<Null>	<Null>

Preparing Input Data for SSNbler and SSN2

7) Assign weighting variable to all points

**NHDPlus and NSI =
TotDASqKM**

Use spatial join

	FID	Shape *	OBSPRED_ID	SITE_ID	SampleYear	MeanAugT	CUMDRAINAG
28	27	Point	28	11548	2000	11.640968	46.64
29	28	Point	29	39218	2014	8.787097	46.53
30	29	Point	30	39244	2009	7.94871	44.21
31	30	Point	31	39244	2014	7.955806	44.21
32	31	Point	32	39245	2014	8.316129	44.21
33	32	Point	33	39261	2015	10.104516	2.53
34	33	Point	34	39277	2013	8.899032	44.21
35	34	Point	35	39277	2014	8.546129	44.21
36	35	Point	100000	0	0	0	6.11
37	36	Point	100001	0	0	0	6.11
38	37	Point	100002	0	0	0	6.11
39	38	Point	100003	0	0	0	6.11
40	39	Point	100004	0	0	0	2.95
41	40	Point	100005	0	0	0	2.95
42	41	Point	100006	0	0	0	2.95

Preparing Input Data for SSNbler and SSN2

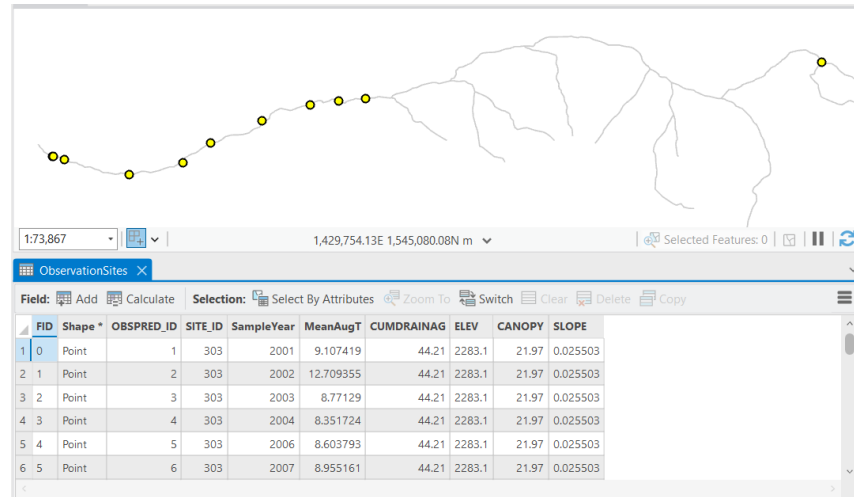
**8) Assign
covariates to
all points**

**9) Split
observations
and
predictions
into separate
feature classes**

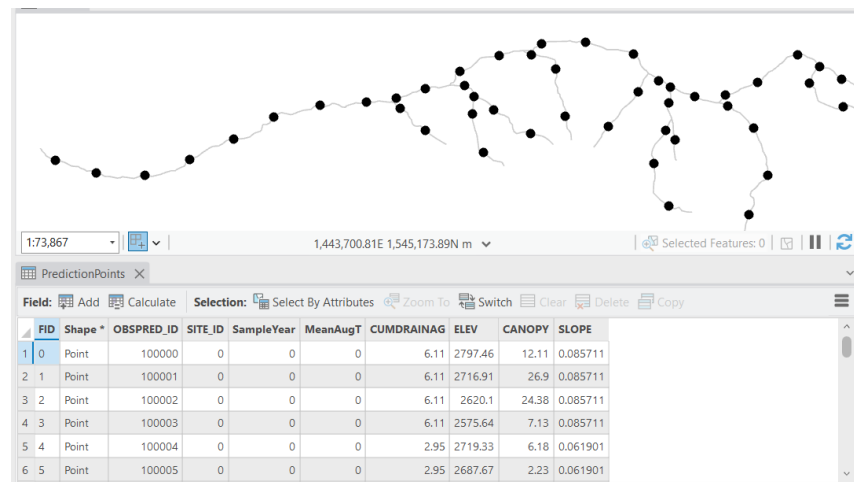
Field:  Add  Calculate		Selection:  Select By Attributes  Zoom To  Switch  Clear  De								
	FID	Shape *	OBSERVED_ID	SITE_ID	SampleYear	MeanAugT	CUMDRAINAG	ELEV	CANOPY	SLOPE
26	25	Point	26	11191	2005	8.333226	44.21	2261.97	32.74	0.025503
27	26	Point	27	11191	2015	9.525484	44.21	2261.97	32.74	0.025503
28	27	Point	28	11548	2000	11.640968	46.64	2071.75	0.71	0.029728
29	28	Point	29	39218	2014	8.787097	46.53	2128	5	0.03441
30	29	Point	30	39244	2009	7.94871	44.21	2261.97	32.74	0.025503
31	30	Point	31	39244	2014	7.955806	44.21	2261.97	32.74	0.025503
32	31	Point	32	39245	2014	8.316129	44.21	2232.36	24.52	0.025503
33	32	Point	33	39261	2015	10.104516	2.53	2750.7	22.33	0.098492
34	33	Point	34	39277	2013	8.899032	44.21	2203.84	29.97	0.025503
35	34	Point	35	39277	2014	8.546129	44.21	2203.84	29.97	0.025503
36	35	Point	100000	0	0	0	6.11	2797.46	12.11	0.085711
37	36	Point	100001	0	0	0	6.11	2716.91	26.9	0.085711
38	37	Point	100002	0	0	0	6.11	2620.1	24.38	0.085711
39	38	Point	100003	0	0	0	6.11	2575.64	7.13	0.085711
40	39	Point	100004	0	0	0	2.95	2719.33	6.18	0.061901
41	40	Point	100005	0	0	0	2.95	2687.67	2.23	0.061901
42	41	Point	100006	0	0	0	2.95	2623.02	2.04	0.061901

Preparing Input Data for SSNbler and SSN2

Observations sites



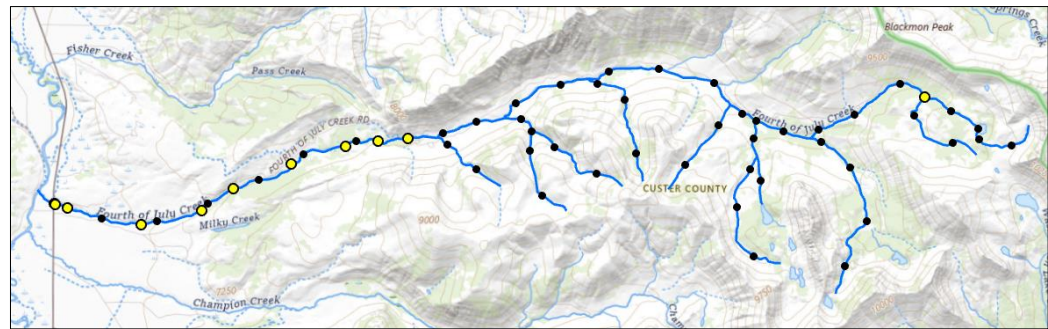
Prediction points



Preparing Input Data for SSNbler and SSN2

Preparing Input Data for Spatial Stream Network Modeling

ObservationSites.shp
PredictionPoints.shp



10) Import sites and preds into LSN, as .gpkg or .shp



An Introduction to SSNbler: Assembling Spatial Stream Network (SSN) Objects in R

```
library(sf)
MF_streams <- st_read(paste0(path, "/MF_streams.gpkg"))
MF_obs <- st_read(paste0(path, "/MF_obs.gpkg"))
MF_pred1km <- st_read(paste0(path, "/MF_pred1km.gpkg"))
MF_CapeHorn <- st_read(paste0(path, "/MF_CapeHorn.gpkg"))
```

Link: <https://pet221.github.io/SSNbler/articles/introduction.html>

Addendum I: GeoPackage does not display in ArcGIS Pro

Problem: Pro needs a projection parameter called “Well Known ID” (WKID) in the projection parameters. Streamlines, observation points, pred points.



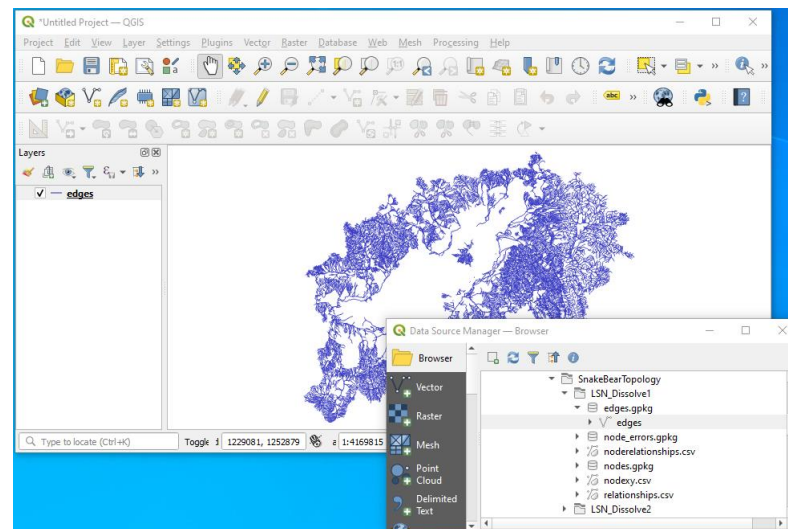
Projected Coordinate System	NAD 1983 Contiguous USA Albers
Projection	Albers
WKID	5070
Authority	EPSG
Linear Unit	Meters (1.0)
False Easting	0.0
False Northing	0.0
Central Meridian	-96.0
Standard Parallel 1	29.5
Standard Parallel 2	45.5
Latitude Of Origin	23.0

Geographic Coordinate System	NAD 1983
WKID	4269
Authority	EPSG
Angular Unit	Degree (0.0174532925199433)
Prime Meridian	Greenwich (0.0)
Datum	D North American 1983
Spheroid	GRS 1980
Semimajor Axis	6378137.0
Semiminor Axis	6356752.314140356
Inverse Flattening	298.257222101

Addendum I: GeoPackage does not display in ArcGIS Pro

Solution 1: Use QGIS to convert GeoPackage to Shapefile (Pro does not work)

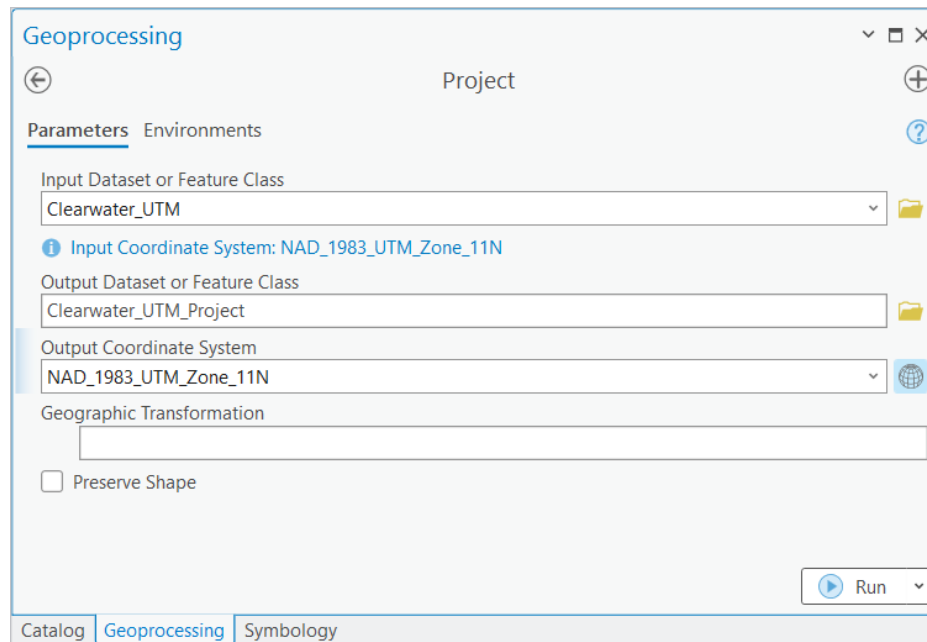
- 1) Display GeoPackage feature class in QGIS
- 2) Right click feature class (edges)
- 3) Export > Save feature class as
- 4) Choose: Esri shapefile



Addendum I: GeoPackage does not display in ArcGIS Pro

Solution 2: Project feature class to a “well-known” coordinate system before generating the LSN

Pro: Data Management Tools > Projections and Transformations > Project



Addendum II: Check Network Topology

Ideally run Check Topology on streamlines before beginning the data preparation steps.

```
## Build the initial LSN and check the topology
lsn_path1<- "E:/SpatialData/Temporary/WKID_Test/LSN2"
edges<- lines_to_lsn(
  streams = river_net,
  lsn_path = lsn_path1,
  snap_tolerance = 1,
  check_topology = TRUE,
  topo_tolerance = 20,
  overwrite = TRUE,
  verbose = TRUE,
  remove_ZM = TRUE)
```



Workflow Tab

<https://research.fs.usda.gov/rmrs/projects/spatial-stream-network#workflow>

Overview Software Requirements **Workflow** Example Data and Other Software People Resources

The following flow chart steps the user through the spatial data processing and spatial statistical modelling process using GIS, SSNbler, and the SSN2 package. A general description of these steps is provided below, along with links to tutorial and example datasets.

1. Data Preparation: Prepare input data including the stream network, observation sites, prediction sites, and covariate (i.e. predictor) data.

[Tutorial](#)

[Example data](#)

[Presentation](#)

2. Topology Correction: Identify and correct topological errors, along with ensuring the topological assumptions required for modeling in SSN2 are met.

[ArcGIS Pro topology tutorial and example data](#)

[QGIS topology tutorial & example data](#)

